Queen Mary
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EBU5376
Digital Signal Processing

Week 3 – Tutorial 1

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Tutorial Lecture 3.1 3.2

Covered in this tutorial:

- Types of DSP Systems
- Difference Equation
- Impulse Response
- Step Response

Types of DSP Systems

- Common types of DSP systems:

1. Linear systems
2. Time-invariant systems
3. Causal / Non-causal / Anti-causal systems
4. Memoryless systems
5. Stable / Unstable / Marginally Stable systems

A system may exhibit more than one properties at the same time

Difference Equation

- Definition: The difference equation expresses the outputs $y[n], y[n-1], \ldots$ of the system as a function of inputs $x[n], x[n-1], \ldots$.

- The general form of a difference equation for an LTI system can be represented as:

$$y[n] + a_1 y[n-1] + \cdots + a_N y[n-N] = b_0 x[n] + b_1 x[n-1] + b_M x[n-M]$$

where $a_1, \ldots, a_N$ and $b_0, \ldots, b_M$ are the coefficients of the difference equation.

- M and N are the memory lengths for input $x[n]$ and output $y[n]$.

Difference Equation

- The general form can be reformulated as

$$y[n] = -a_1 y[n-1] - \cdots - a_N y[n-N] + b_0 x[n] + b_1 x[n-1] + \cdots + b_M x[n-M]$$

Current **output**

Past **outputs**

Current and past **inputs**

- Equivalently,

$$y[n] = -\sum_{i=1}^{N} a_i y[n-i] + \sum_{j=0}^{M} b_j x[n-j]$$

*This is a **recursive** equation.*

Solving Difference Equation with Initial Conditions

- Steps to solve a difference equation via z-transform:
  1. Reformulate the difference equation, with the input and the output on separate sides of the equation
  2. Take the z-transform on both sides
  3. Substitute $Y[z] = Z(y[n]) = y[0] + z^{-1}y[1] + z^{-2}y[2] + \cdots$
     - Use the initial condition to simplify the equation
  4. Apply partial fraction decomposition
  5. Perform inverse z-transform to obtain the solution $y[n] = \cdots$

Impulse Response, Step Response

- If the difference equation is *recursive*,
  - The **impulse response** $h[n]$ is the solution to the difference equation, when the input is the unit impulse function $\delta[n]$
    - It provides a system's natural response to a sudden, unit magnitude input at time zero.
  - The **step response** $s[n]$ is the solution to the difference equation, when the input is the unit step function $u[n]$
    - It shows how the system responds to a sustained change in input.

Question 1, Part 1

- Identify the linear systems from the options below.

  a. $y[n] = x[3n - 2]$ Linear

  b. $y[n] = x[n] + 2x[n - 1]$ Linear

  c. $y[n] = x[n - 2] + 3$ Non-linear

  d. $y[n] = x[-n]$ Linear

Question 1, Part 1 - Answers

- For each system, need to check that it is both homogeneous and additive

System a $y[n] = x[3n-2]$

If the input is $ax[n]$, the output is $ax[3n-2] = ay[n]$. Hence, the system is homogeneous

If the input is $x_1[n] + x_2[n]$, the output is $x_1[3n-2] + x_2[3n-2] = y_1[n] + y_2[n]$. Hence, the system is additive

Therefore, this is a linear system

We can also verify the two properties at once:

System b $y[n] = x[n] + 2x[n-1]$

Let $ax_1[n] + bx_2[n]$ be the input, the output becomes

$$\begin{aligned} & ax_1[n] + bx_2[n] + 2ax_1[n-1] + 2bx_2[n-1] \\ &= a(x_1[n] + 2x_1[n-1]) + b(x_2[n] + 2x_2[n-1]) \\ &= ay_1[n] + by_2[n] \end{aligned}$$

Hence, the system is both homogeneous and linear, thus a linear system.

Both two approaches are correct. The second one may lead to more complicated math derivations.

Question 1, Part 1 - Answers

- For each system, need to check that it is both homogeneous and additive

System c $y[n] = x[n-2] + 3$

If the input is $ax[n]$, the output is $ax[n-2] + 3 \neq ay[n]$. The system is NOT homogeneous

If the input is $x_1[n] + x_2[n]$, the output is $x_1[n-2] + x_2[n-2] + 3 \neq y_1[n] + y_2[n]$. The system is not additive

Therefore, this is NOT a linear system

System d $y[n] = x[-n]$

If the input is $ax[n]$, the output is $ax[-n] = ay[n]$. The system is homogeneous

If the input is $x_1[n] + x_2[n]$, the output is $x_1[-n] + x_2[-n] = y_1[n] + y_2[n]$. The system is additive

Therefore, this is a linear system

Question 1, Part 2

- Identify the time-invariant systems from the options below.

a. $y[n] = x[3n - 2]$	Time-varying
b. $y[n] = x[n] + 2x[n - 1]$	Time-invariant
c. $y[n] = x[n - 2] + 3$	Time-invariant
d. $y[n] = x[-n]$	Time-varying

Question 1, Part 2 - Answers

- For each system, check whether $y[n-m]$ is the output of $x[n-m]$

<u>*System a*</u> $y[n] = x[3n-2]$

If the input is delayed by m steps, i.e., $x[n-m]$, the output would be

$$x[3n-2-m]$$

If the output is delayed by m steps, i.e., $y[n-m]$, the value of the output is

$$x[3n-2-3m]$$

Since $y[n-m]$ is not the output of $x[n-m]$, the system is NOT time-invariant

Question 1, Part 2 - Answers

- For each system, check whether $y[n-m]$ is the output of $x[n-m]$

<u>*System b*</u> $y[n] = x[n] + 2x[n-1]$

Let $x[n-m]$ be the input, the output is $x[n-m] + 2x[n-1-m]$

We have $y[n-m] = x[n-m] + 2x[n-m-1]$, which is equal to the output of $x[n-m]$

Hence, the system is time-invariant

<u>*System c*</u> $y[n] = x[n-2] + 3$

Let $x[n-m]$ be the input, the output is $x[n-2-m] + 3$

We have $y[n-m] = x[n-m-2] + 3$, which is equal to the output of $x[n-m]$

Hence, the system is time-invariant

<u>*System d*</u> $y[n] = x[-n]$

Let $x[n-m]$ be the input, the output is $x[-n-m]$

We have $y[n-m] = x[-(n-m)]$, which is not equal to the output of $x[n-m]$

Hence, the system is NOT time-invariant

Question 1, Part 3

- Identify the memoryless systems from the options below.

  a. $y[n] = x[3n - 2]$

  b. $y[n] = x[n] + 2x[n - 1]$

  c. $y[n] = x[n - 2] + 3$

  d. $y[n] = x[-n]$

None.
All of these systems need a memory unit.

Question 1, Part 4

- Identify the linear time-invariant (LTI) systems from the options below.

a. $y[n] = x[3n - 2]$	Linear but not time-invariant
b. $y[n] = x[n] + 2x[n - 1]$	LTI
c. $y[n] = x[n - 2] + 3$	Non-linear but time-invariant
d. $y[n] = x[-n]$	Linear but not time-invariant

Question 2

- Derive the **impulse response** of the following system

$$y[n] = 0.16y[n-2] + x[n], \qquad y[-1] = 0, y[-2] = 0$$

The challenge is to determine the relationship between $Z(y[n])$ and $Z(y[n-2])$ for this system with the given initial conditions.

To illustrate this, we first write out the two sequences:

$$y[n] = [y[0], y[1], y[2], \dots], n \geq 0$$

$$y[n-2] = [y[-2], y[-1], y[0], y[1], \dots], n \geq 0$$

The z-transform of them is given by

$$Z(y[n]) = y[0] + z^{-1}y[1] + z^{-2}y[2] + \cdots$$

$$Z(y[n-2]) = y[-2] + z^{-1}y[-1] + z^{-2}y[0] + z^{-3}y[1] + z^{-4}y[2] + \cdots$$

Therefore, the relationship between the two z-transform is

$$Z(y[n-2]) = z^{-2}Z(y[n]) + z^{-1}y[-1] + y[-2]$$

Using the initial conditions, the equation is simplified to

$$Z(y[n-2]) = z^{-2}Z(y[n])$$

Question 2

- Derive the **impulse response** of the following system

$$y[n] = 0.16y[n-2] + x[n], \qquad y[-1] = 0, y[-2] = 0$$

Based on the previous equation, we derive the impulse response

First rearrange the terms in the difference equation and let $x[n] = \delta[n]$, we get

$$y[n] - 0.16y[n-2] = \delta[n]$$

Apply z-transform, substitute $Z(y[n-2]) = z^{-2}Z(y[n])$, and denote $Z(y[n]) = Y(z)$:

$$Y(z) - 0.16z^{-2}Y(z) = 1$$

$$Y(z) = \frac{1}{1 - 0.16z^{-2}}$$

By applying partial fraction,

$$Y(z) = \frac{1}{2} \cdot \frac{1}{1 + 0.4z^{-1}} + \frac{1}{2} \cdot \frac{1}{1 - 0.4z^{-1}}$$

By inverse z-transform,

$$y[n] = 0.5 \cdot (-0.4)^n u[n] + 0.5(0.4)^n u[n]$$

This is the system response when the input is $\delta[n]$. Hence, it is the impulse response, i.e.,

$$h[n] = 0.5 \cdot (-0.4)^n u[n] + 0.5(0.4)^n u[n]$$

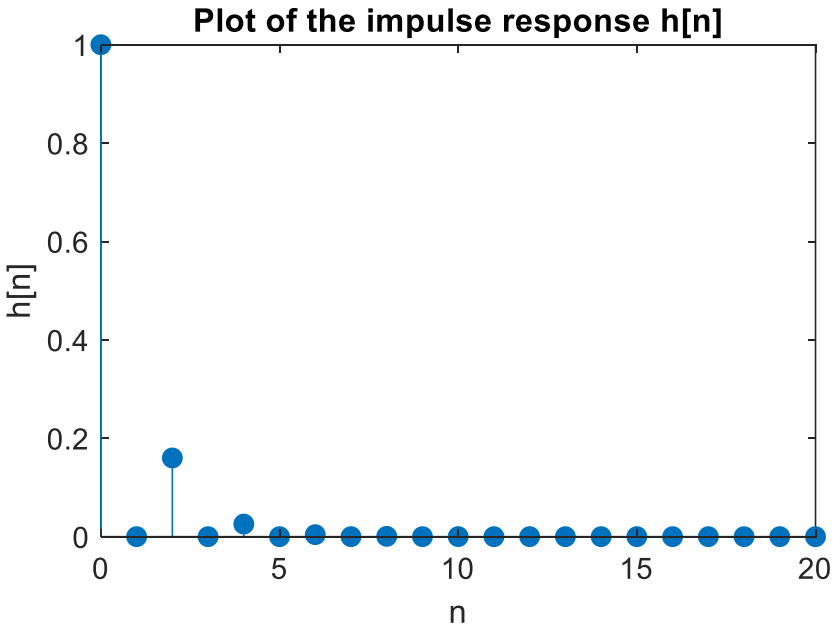
Question 2

- Derive the **impulse response** of the following system

$$y[n] = 0.16y[n-2] + x[n], \qquad y[-1] = 0, y[-2] = 0$$

Here is a plot of the impulse response for $n = 0, \dots, 20$

The impulse response shows that the system quickly responds to a sudden input and then stabilizes after a few fluctuations.



Question 3

- Derive the **step response** of the following system

$$y[n] = 0.5y[n-1] + x[n], \qquad y[-1] = 1$$

Rearrange the terms in the difference equation and let $x[n] = u[n]$, we get

$$y[n] - 0.5y[n-1] = u[n]$$

By applying z-transform, we get

$$Z(y[n]) - 0.5Z(y[n-1]) = \frac{1}{1-z^{-1}}$$

Since

$$Z(y[n]) = y[0] + z^{-1}y[1] + z^{-2}y[2] + \cdots$$
$$Z(y[n-1]) = y[-1] + z^{-1}y[0] + z^{-2}y[1] + z^{-3}y[2] + \cdots$$

We drive the relationship

$$Z(y[n-1]) = z^{-1}Z(y[n]) + y[-1]$$

Substitute the initial condition and denote $Y(z) = Z(y[n])$, the z-transform becomes

$$Y(z) - 0.5z^{-1}Y(z) - 0.5 = \frac{1}{1-z^{-1}}$$

Question 3

- Derive the **step response** of the following system

$$y[n] = 0.5y[n-1] + x[n], \qquad y[-1] = 1$$

By rearranging the terms, the previous equation becomes

$$Y(z) = \frac{1.5 - 0.5z^{-1}}{(1 - z^{-1})(1 - 0.5z^{-1})}$$

By partial fraction, we get

$$Y(z) = \frac{2}{1 - z^{-1}} - \frac{0.5}{1 - 0.5z^{-1}}$$

By inverse z-transform, the solution to this difference equation is

$$\begin{aligned} y[n] &= 2u[n] - 0.5 \cdot (0.5)^n u[n] \\ &= 2u[n] - (0.5)^{n+1} u[n] \end{aligned}$$

Which is the step response of this system, i.e.,

$$s[n] = 2u[n] - (0.5)^{n+1} u[n]$$

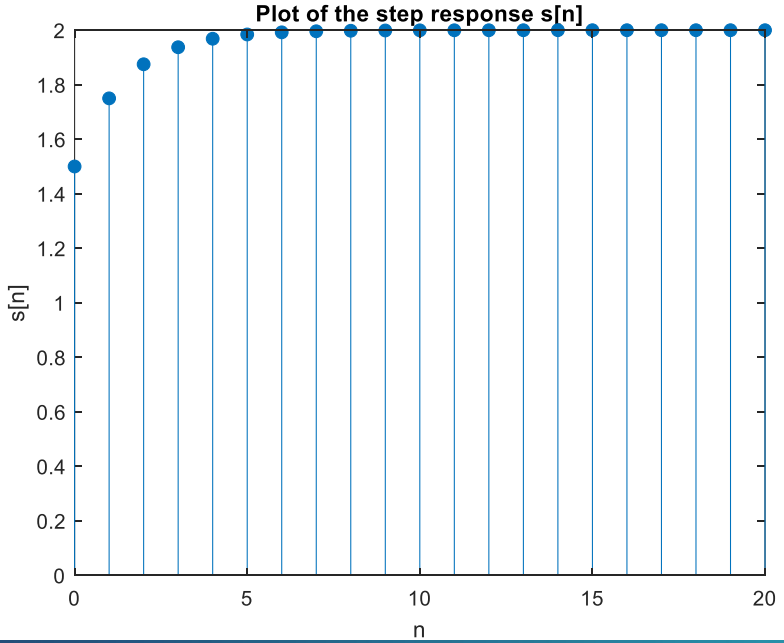
Question 3

- Derive the **step response** of the following system

$$y[n] = 0.5y[n-1] + x[n], \qquad y[-1] = 1$$

Here is a plot of the step response for $n = 0, \ldots, 20$

The step response shows that the system responds slowly to a change in input, by approaching the steady state after ~5 time steps.



Question 4

- In this question, you'll be presented with various practical scenarios requiring analysis of DSP system responses.

- Decide whether to use **impulse or step response** for each situation, and briefly explain why it's the preferred choice

Question 4

- **Scenario 1**: Designing a noise-cancellation headphone system requires swift reaction to various sound levels. Which time-domain response would guide us in verifying system reactivity?

The impulse response would be the appropriate option. It shows the system's immediate reaction to brief, sharp sounds, which is critical for effective noise cancellation.

Question 4

- **Scenario 2**: A smart home thermostat is tested for its response to user-adjusted temperature settings. To ensure comfort, which time-domain response would best reflect the system's ability to reach and hold a new temperature?

Step response is preferred.
A thermostat's primary function is to transition smoothly from one temperature to another and hold steady—exactly what a step input tests.

Question 4

- **Scenario 3**: When testing for the airbag's deployment timing and precision, which time-domain response is essential to consider?

Impulse response is vital as it indicates the airbag system's promptness and accuracy in response to collision-like signals.

Question 5

- You're building an audio compressor that processes input samples to reduce sudden loud peaks. You consider the following system:

$$y[n] = \max(x[n], x[n-1])$$

Which of the following properties does this satisfy?

- Linearity
- Time-invariance
- Causality
- Memory

Briefly justify your answer.

Question 5

- Linearity

The max function is non-linear.

If input is $ax[n]$, the output becomes $\max(ax[n], ax[n-1])$ which is not always equal to $a\max(x[n], x[n-1])$.
For example, if $x[n] > x[n-1]$ and $a < 0$, we have $ax[n] < ax[n-1]$
Hence the system is not homogeneous

For additivity, if input is $x_1[n] + x_2[n]$ then:

$$y[n] = \max(x_1[n] + x_2[n], \quad x_1[n-1] + x_2[n-1])$$
$$\neq \max(x_1[n], x_2[n]) + \max(x_1[n-1], x_2[n-1])$$

So the system does not satisfy additivity either.

Question 5

- Time invariance

The system is time-invariant.

If input is delayed by k, i.e., $x[n-k]$, the output is $\max(x[n-k], x[n-k-1])$ which is exactly the same as $y[n-k]$

- Causality

The system is causal.

The output at time n depends only on the present ($x[n]$) and past ($x[n-1]$) inputs. No future inputs are used, so this system is causal and can be implemented in real-time.

- Memory

Since $y[n]$ depends not just on the current input $x[n]$ but also on a past value $x[n-1]$, the system has memory.

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